



MATHEMATICS

GRADE ONE-TERM ONE



SECTION ONE

Introduction

Mathematics is a subject that deals with numbers and symbols whose relationship is expressed in rules. Mathematics enables the learner to acquire competencies, values and positive attitudes which enable them to develop logical thinking, ability to apply the knowledge acquired, analyze situations and make rational decisions. The competencies that learners acquire enable them to continue with further education, lead a productive life and contribute to National economic development.

The subject is organized such that the content taught in one level becomes the prerequisite in the next level. The content therefore becomes more demanding as we move from one level to another.



VISION : an **engaged, empowered** and **ethical** citizen

MISSION: Nurturing every learner's potential

National Goals of Education

Education in Kenya should:

1. Foster nationalism, patriotism, and promote national unity

Kenya's people belong to different communities, races and religions and should be able to live and interact as one people. Education should enable the learner acquire a sense of nationhood and patriotism. It should also promote peace and harmonious co-existence.

2. Promote social, economic, technological and industrial needs for national development

Education should prepare the learner to play an effective and productive role in the nation.

a) Social Needs

Education should instill social and adaptive skills in the learner for effective participation in the family, community, national, regional and international development.

b) Economic Needs

Education should prepare a learner with requisite competences that support a modern and independent growing economy. This should translate into high standards of living for every individual.

c) Technological and Industrial Needs

Education should develop in the learner necessary competences for technological and industrial development for the nation in tandem with global trends.

3. Promote individual development and self-fulfillment

Education should provide opportunities for the learner to develop to the fullest potential. This includes development of one's interests, talents and character for positive contribution to the society.



4 Promote sound moral and religious values

Education should promote acquisition of national values as enshrined in the Kenya Constitution. It should be geared towards developing a self-disciplined and ethical citizen with sound moral and religious values.

- 5. Promote social equity and responsibility** Education should promote social equity and responsibility. It should provide inclusive and equitable access to quality and differentiated education; including for learners with special educational needs and disabilities. Education should also provide the learner with opportunities to develop and practice shared responsibility and accountability through community service learning.

6. Promote respect for and development of Kenya's rich and varied cultures

Education should instill in the learner appreciation of Kenya's rich and diverse cultural heritage. The learner should value own and respect other people's culture as well as embrace positive cultural practices in a dynamic society.

7. Promote international consciousness and foster positive attitudes towards other nations

Kenya is part of the interdependent network of diverse peoples and nations. Education should empower the learner to respect, appreciate and participate in the opportunities within the international community. Education should also enable the learner to operate within the international community with full knowledge of the obligations, responsibilities, rights and benefits that this membership entails.

8. Promote positive attitudes towards good health and environmental protection

Education should inculcate in the learner the value of physical and psycho-social well-being for self and others. It should promote environmental preservation and conservation, including animal welfare, for sustainable development.



Learning Outcomes for Early Years Education

By the end of early years" education, the learner should be able to:

1. Demonstrate basic literacy and numeracy skills for learning.
2. Communicate appropriately using verbal and/or non-verbal modes in a variety of contexts.
3. Demonstrate appropriate etiquette in social relationships.
4. Apply creativity and critical thinking skills in problem solving.
5. Explore the immediate environment for learning and enjoyment.
6. Practice hygiene, nutrition, sanitation, safety skills to promote health and wellbeing.
7. Demonstrate the acquisition of emotional, physical, spiritual, aesthetic and moral development for balanced living.
8. Demonstrate appreciation of the country's rich and diverse cultural heritage for harmonious co-existence.
9. Apply digital literacy skills for learning and enjoyment.

INTERRELATIONSHIP BETWEEN NATIONAL GOALS OF EDUCATION, LEVEL LEARNING OUTCOMES, SUBJECT GENERAL LEARNING OUTCOMES, SPECIFIC LEARNING OUTCOMES AND THE LEARNING EXPERIENCES

The level learning outcomes for early years education are derived from the National goals of education. It is from the early years education level learning outcomes that the General learning outcomes for mathematics activities are derived. The specific learning outcomes are then derived from the general learning outcomes from which the learning experiences are formulated.

The following table shows the relationship.

NATIONAL GOALS OF EDUCATION	LEVEL LEARNING OUTCOMES	SUBJECT GENERAL LEARNING OUTCOMES	SPECIFIC LEARNING OUTCOMES	SUGGESTED LEARNING EXPERIENCES
Foster nationalism, patriotism, and promote national unity	Communicate appropriately using verbal and/or non-verbal modes	Describe properties of Geometrical shapes and spatial relationships	Identify rectangles, circles and triangles in the environment	Learners in pairs /groups discuss the types of lines that make rectangles, circles, triangles



NATIONAL GOALS OF EDUCATION	LEVEL LEARNING OUTCOMES	SUBJECT GENERAL LEARNING OUTCOMES	SPECIFIC LEARNING OUTCOMES	SUGGESTED LEARNING EXPERIENCES
	in a variety of contexts.	in real life experiences		and name them
Promote social, economic, technological and industrial needs for national development	Demonstrate basic literacy and numeracy skills for learning.	Demonstrate mastery of Number concepts by working out problems in day to day life.	Subtract a 1-digit number from a 2- digit numbers based on basic addition facts	Learners to solve routine and non-routine problems involving subtraction of a 1-digit number from a 2- digit numbers based on basic addition facts
Promote positive attitudes towards good health and environmental protection	5. Explore the immediate environment for learning and enjoyment.	Demonstrate mastery of Number concepts by working out problems in day to day life.	Sort and group objects according to different attributes within the classroom	Learners in pairs/groups to sort objects with same attribute and group them together



CORE COMPETENCIES

There are seven (7) core competencies to be achieved by every learner in basic education:

1. Communication and Collaboration
2. Critical Thinking and Problem Solving
3. Creativity and Imagination
4. Citizenship
5. Digital Literacy
6. Learning to Learn
7. Self-efficacy

Communication and collaboration

Communication is the act of transferring information from one place to another, whether vocally, visually, or non-verbally.

Collaboration: Collaboration is the process of two or more people or organizations working together to realize shared goals. Collaboration may require leadership, social within a decentralized or egalitarian group teams that work collaboratively in relation to gaining greater resources, recognition and motivation.

Self-efficacy

Self-efficacy is the person's belief about his or her capabilities to perform tasks or assignments that can change and transform his or her life. It determines how the person feels, thinks, behaves and motivates self positively. Self-efficacy has the potential to determine four major processes namely cognitive, motivational, and affective and selection processes.

Self-efficacy will enable learners to develop and nurture intra-personal skills and values such as self-awareness, self-esteem, confidence and personal integrity.

Critical thinking and problem solving

There are three types of thinking: reasoning, making judgments, and problem solving. When learners are empowered with critical thinking, they avoid being subjective, and use logic and evidence to arrive at conclusions. Critical thinking also facilitates exploring new ways of doing things and learner autonomy.



Critical thinking is important for lifelong learning. It helps learners to have an open mind and be ready to listen and appreciate information and opinions that may sometimes conflict with their earlier held beliefs and positions. Critical thinking and problem solving are useful for learners of all ages and in all subjects and disciplines offered in the basic education curriculum. For example, in the Sciences learners need to think critically about observations and patterns to develop ideas on how to solve problems. The competences are also important for solving problems in their lives and community, and ultimately achieve their potential which is the vision for basic education curriculum.

Creativity and Imagination

Imagination and Creativity refers to the ability to form new images and sensations in the mind, and to turn them into reality. It is the ability to imagine things that are not real, to form pictures in the mind, of things that one has not seen or experienced, and turning those pictures into real things. Imagination and Creativity on the other hand, is characterized by ability to perceive the world in new ways, to find hidden patterns, to make connections between seemingly unrelated phenomena, and to generate solutions.

In an educational set up, imagination and creativity refers to the ability of learners and their teachers to form images and ideas in their minds, and turn them into real, visible creations. Learners who are imaginative and creative are able to make life interesting for themselves and others around them. They are able to use the knowledge, skills and values acquired in the learning process to create new ideas which result in products that add value to their lives and to those of others around them.

Citizenship

Citizenship is the state of being vested with the rights, privileges, and duties of a citizen. It creates a sense of belonging and attachment to one's nation. A sense of citizenship helps to equip young people to deal with situations of conflict and controversy knowledgeably and tolerantly. They are able to understand the consequences of their actions, and those of the adults around them.



Digital literacy

Digital literacy is the knowledge, skills and behaviour in a broad use of digital content and devices. Such devices include mobile phones, smartphones, iPhone, tablets, laptops and desktops among others. All these are within the category of network devices. Digital literacy therefore focuses mainly on networking devices and should not be confused with computer literacy skills. However, traditional forms of literacy and computer literacy are enhancers in the acquisition of digital literacy skills.

Learning to learn

Learning to learn is the ability to pursue and persist in learning, to organize one's own learning, by effective management of time and information, both individually and in groups. This competence includes awareness of one's learning process and needs, identifying available opportunities, and the ability to overcome obstacles in order to learn successfully. This competence means gaining, processing and assimilating new knowledge and skill as well as seeking and making use of guidance. Learning to learn helps learners to build on prior learning and life experiences in order to use and apply knowledge and skills in a variety of contexts. There are four pillars of learning: Learning to know, learning to do, learning to be and learning to live together.

VALUES-BASED EDUCATION

Introduction

Values are standards that guide people on how to respond or behave in each situation. They influence how someone feels, acts and makes choices in life.

The responsibility for **nurturing values** rests with parents, school and the community. Everybody in the school has a role to play in helping the learner to nurture values. This approach is referred to as 'The whole school approach to value based education'.

The overall goal of values-based education is 'To nurture values in learners to become empowered, engaged and ethical citizens for positive and holistic transformation of society'. The teacher therefore has an important role to play in nurturing values through teaching.



Core Values

According to the competency based Curriculum there are eight core values to be nurtured in the learner namely love, responsibility, respect, unity, peace, patriotism, social justice, patriotism integrity **Refer to the appendix... Core Values.** The teacher is expected to identify value(s) to be nurtured in the learning experience in every strand. He/she is expected to constantly guide and engage the parents to reinforce the values learners have learnt in school at their home. Learners are expected to emulate exemplary behaviours and values from teachers, family members and the community at large hence the need for adults to exhibit good behaviour. Learners spend most of their formative years in school, which presents opportunities for the teacher to mould and reinforce values upon which the learner's character is formed.

PERTINENT AND CONTEMPORARY ISSUES

Introduction

Holistic, meaningful and learner centered education does not only focus on the subject content but also on the competencies the learner requires to effectively address the issues and challenges of everyday life thereby leading a fulfilled life and becoming a productive member of the society. Issues that have been identified as salient and therefore mainstreamed in the designs are referred to as Pertinent and contemporary issues (PCIs). Mainstreaming is the process of identifying suitable opportunity in the learning area or subject where PCI can be incorporated appropriately and taught alongside the subject matter. PCIs should be included in the schemes of work and in the lesson plans.

Classification of PCIs

PCIs have been classified into six broad areas comprising of several related salient issues as follows:

1. **Citizenship.** Peace Education, Integrity, ethnic and racial relations, social cohesion, patriotism, good governance, child's rights, child care and protection,
2. **Health Education Panel,** HIV and AIDS Education, Alcohol and drug abuse prevention, Life style diseases, personal hygiene, Common Communicable and non-communicable diseases and chronic, diseases
3. **Life Skills and Values Education:** Life Skills Education, Core Values and Human sexuality.
See the appendices on Life Skills Education and Core values



4. **Education for Sustainable Development (ESD):** Environmental Education, Disaster Risk Reduction, Safety and security Education, Financial Literacy, Poverty eradication, Countering terrorism, extreme violence and radicalization, Gender issues in education and Animal Welfare Education
5. **Learner Support Programmes (Non Formal Education programmes):** Guidance services, Career guidance, counselling services, Peer education, mentorship, chaplaincy services, clubs and societies, Sports and games.
6. **Parental Empowerment and engagement and, Community Service learning:** Parental empowerment and engagement and, Community and Service learning.

During lesson preparation, the teacher has to be deliberate in identifying specific PCIs and planning how they shall be incorporated in the lesson. He /She should ensure activities on PCIs are undertaken and covered during the learning experiences.

The teacher is expected to constantly guide and engage the parents to reinforce what the learner has learnt in school at home. The school community and the larger community should provide conducive environment for learners to apply competencies they have learnt on PCIs to address issues at home and in the society. Please refer to the sample lesson plan given in the handbook/Teachers guide.

PCIs are supposed to not only be integrated in the lesson but in other programmes and activities in the school, home and in the community. Acquisition of knowledge, values and skills in various PCIs will enable the learner to translate what they have learnt to real life situation both in and outside school.

Differentiated Learning

Every learner learns differently. Technically, individual's learners have preferential way in which they absorb, process, comprehend and retain information. It is therefore important for educators to understand the differences in their learners' preferential way of learning, so that they can implement best practice strategies into their daily activities, curriculum and assessments.



Differentiated Learning is effective instruction that is responsive to learners' readiness, interests and learning preferences. It enables learners to build new learning through connections to existing knowledge and preferred ways of learning. Readiness is the learner's starting point for learning, relative to the concept being studied while his or her interests enhances the relevancy of learning by linking new information to his or her experience and enthusiasm. The learning preferences refer to different ways in which learners prefer to acquire, process and work with information.

Learning preferences are influenced by gender, culture, the classroom environment, learning styles and multiple intelligences. Teachers are required to use a range of instructional and assessment strategies to meet the needs of different learners. Learners are assessed before, during and after learning. Assessment informs next steps for both teacher and learner. All differentiated instruction activities are equally engaging and respectful and take approximately the same amount of time.

Differentiated learning allows for flexibility in creating and altering instructional plans in response to learners. It involves expanding instructional routines and skills where learning experiences are introduced to learners in different ways of learning. Differentiated learning provides effective teaching that involves providing different learners with different avenues to learning.

USE OF INFORMATION COMMUNICATION TECHNOLOGY (ICT)

ICT entails the use of technology in information delivery. Consequently ICT in education entails teaching and learning using technology devices and material.

The resources used in ICT are mainly digital, electronic and Portable Document Format (PDF) files. The bulk of ICT education materials in KICD are radio, television, e-learning programmes and PDFs.

- a) **Radio programmes** - involve audio recordings disseminated through radio/electronic devices
- b) **Television (Video) programmes** - audio-visual content that is mainly video based and disseminated through television and other electronic devices.



- c) **E – learning (computer based) programmes** - electronic learning materials disseminated through computer /smart T.Vs technology.
- d) **PDFs** - electronically converted images from print material.

The education ICT content is stored in various electronic formats including offline modes such as CDs, DVDs, Flash disks, Memory cards and online formats.

The KICD electronic content can be accessed from the following Digital Literacy Programme devices in the schools:

- Teacher Digital Device (TDD)
- Learner Digital Device (LDD)
- Digital Content Server & Wireless Router (DCSWR)
- External Hard Disk

When chosen and used appropriately, the electronic material will enable the learner use modern technologies thus making ICT technology a delivery tool for all learning areas. This added advantage to teaching and learning enhances learning by;-

- i. Supplementing classroom instruction
- ii. Stimulating and motivating the learner
- iii. Enhancing concepts acquisition
- iv. Arousing learners interest and promoting active participation during the lesson
- v. Saving time used to explain concepts
- vi. Enhancing skills development
- vii. Enabling the teacher to cater for individual differences
- viii. Reaching learners by multisensory presentation

Research findings indicate that retention of knowledge and skills is highly accomplished through the three sensory domains of audio, audio-visual and tactile. The teachers are therefore encouraged to use quality Educational Resources. The endeavor will enable teachers to access resources that will address the needs of the learner in order to make learning interesting and more meaningful. When choosing the appropriate resource, the teacher should take note of the following;-



- a) Relevance to the content being taught
- b) Accuracy
- c) Stimulation of learner's imagination and enhancement of concepts acquisition
- d) Appropriateness to the level of the learner
- e) Capability of capturing the learner's interest for active participation
- f) Durability and safety for learners use
- g) Application of new knowledge and skills
- h) Conformity to set standards
- i) Awareness creation to teachers on the latest pedagogical practices
- j) Interactivity and sustainability

The use of technology to teach and learn is a major educational reform.

It can be used as a tool for learning. It is a valuable resource which can be used:

- ✓ As a method for teaching
- ✓ For assessment
- ✓ To introduce the lesson and
- ✓ To enhance what was learnt

Information and Communication Technologies play an increasingly important role in the way we communicate, learn and live. The challenge is to effectively harness these technologies in a way that serves the interests of learners in their learning environment.

ICT should contribute to access to education, equity in education and the delivery of quality learning and teaching. It is not meant to replace the teacher. Instead, consideration must be given to the subject matter, the learning objectives and outcomes, the characteristics of the learners, and the learning context in order to arrive at the optimum mix of instructional and delivery methods. Moreover, different technologies are typically used in combination rather than as the sole delivery mechanism.

PARENTAL EMPOWERMENT AND ENGAGEMENT

Children start learning before they go to school. Parents, other family members and care-givers are a child's first and most important educator. They can be a positive influence and help child



do well at school. Parents therefore require empowerment and opportunity to get involved throughout their children's schooling.

As soon as children enrol for formal learning, teachers have a responsibility to create awareness and provide an enabling environment for parents to engage in their child's learning. Involving parents can have very positive impact on education outcomes. The following are suggested ways teachers can employ to empower and engage parents.

Parents can be sensitized on:

1. Parental roles and responsibilities with regard to education according to the constitution and other national and school policies.
2. The need to ensure the child is given proper care and protection to experience normal patterns of growth and development.
3. The importance of providing basic care, nutrition, clothing, shelter and access to good health.
4. Importance of early identification of disability and prompt referral for early intervention to mitigate the disability in order to avoid further delay or distorted development.
5. The need to have a safe and conducive environment to enable a child to be creative and innovative.
6. Importance of creating a family culture where values taught in school are reinforced at home.
7. The need to equip children with basic self-help skills e.g. toilet training, eating and table manners/etiquette and dressing up.
8. The fact that children need to develop fine motor skills so they can do activities such as scribbling, painting, clothing and opening buttons.
9. The importance of providing resources e.g. school uniform, toys and other items required by the teacher in various learning areas and subjects.
10. The importance of carrying out roles and tasks assigned by the teacher for instance reinforcing a skill, value or provision of extra information on an issue.
11. Reading and narrating to children stories.
12. Importance of seeking information from the school head and teachers about the formal, informal and the non-formal aspects of the child's potential at school.



Parents play a very important role in determining the success of a child's education. Teachers in liaison with the head teacher should from time to time have forum to discuss strategies they can employ to empower and engage parents in their school activities.

Professional documents and their use

In order to plan for effective delivery of the curriculum, it is crucial that teachers plan their work well. Professional documents are used to organise curriculum implementation. Kenya Institute of Curriculum Development develops **curriculum designs**. These are curriculum document which usually contains specific information about the course such as, details on the expected learning outcomes, what is to be taught, how it can be taught, and suggestions for assessment, resources and time allocation. It is therefore a vital document that the teacher must use in the teaching and learning process. The teacher should use the curriculum designs while preparing schemes of work and lesson plans.

Schemes of Work are developed by the implementing teacher to organise how the curriculum will be implemented in a given year. It is developed from the curriculum designs. It helps the teacher distribute what is to be covered within the available time and how it shall be done. It therefore shows how the planned curriculum shall be distributed or organised within the time allocated. The teacher who is expected to teach the activity area should develop the scheme of work as he/she can easily contextualise what is feasible for implementation in his/her context. He or she is able to tell what resources are available for him or her to use in the learning process. A sample Scheme of Work (SoW) is appended at the back of this book.

Lesson plans are developed from the schemes of work. They describe the actual teaching and learning experiences that the teacher will facilitate during the lesson. It provides clear instructions on how the class will be organised, the resources that will be used, and the activities that will take place within the time allocated. It also shows how values, PCIs and competencies shall be mainstreamed in the lesson. Lesson plans are usually propositions made by the teacher on how the lesson shall run, however, the way in which the lesson may run could change spontaneously during the lesson. Lesson plans should be developed by the teacher who will facilitate the lesson as they are better placed to know the type of learners they have, the resources



available to them and the context in which the learning shall take place. A sample lesson plan for a competency based curriculum is provided in the appendix of this book.

After facilitating a lesson the teacher should indicate what was covered in the lesson and how well the learners achieved the learning outcomes envisaged. This document is called the **record of work book**. The record of work book is important as it indicates what the learners have achieved within a given time. It also enables the school management to track what is happening in the classes in their schools. In case the teacher is absent or leaves the school, the record of work book enables the replacing teacher to pick up from where they left and effectively continue implementing the curriculum. Record of work books should be signed by the teachers and also regularly signed by the school management.

In order to track learner's performance the teacher should also have an **assessment record book**. The book will track the achievement of learning outcomes and competencies acquired by the learners. It will guide the teacher on which learner needs special attention to acquire the skills and competencies stated in the curriculum designs. The information will also be used when reporting learner's progress to their parents. A sample of an assessment report is also provided in the appendix of this book.

HOW TO USE THIS HANDBOOK

This handbook contains the following:

- Strands
- Sub-strands
- Specific learning outcomes
- Suggested learning experiences
- Issues in the reformed curriculum
- Assessment

STRANDS AND SUB- STRANDS

This handbook has discussed the concepts in term one of Early Grade Mathematics (E.G.M) under PRIEDE project. The handbook guides the teacher on how to integrate issues in the reformed curriculum with those in EGM.



No.	Strand	Sub-strand
1.	Numbers	Number concept
		Whole Numbers
		Addition
		Subtraction
2.	Measurement	Length
		Time
3.	Geometry	Shapes

In the suggested learning experiences the following issues have been discussed:

- Core competencies
- Pertinent and contemporary issues
- Values
- Links to other subjects
- Suggested non-formal activities to support learning
- Suggested community service learning
- Suggested assessment

The teacher should refer to the key inquiry questions stated under each sub-strand in the curriculum design during the teaching and learning process. On assessment the teacher should always refer to the assessment rubrics stated in the designs as well as the suggested methods indicated in the designs under suggested assessment.

A report card template has been included. This template will assist the teacher in reporting the learner's performance both in relation to the achievement of content in the classroom as well as the achievement of the core competencies.



SECTION TWO

ESSENCE STATEMENT

Numeracy is a foundational skill that prepares the learner for number work, mathematics in higher levels of schooling and mathematical approaches in all aspects of life. Numeracy activities involve identification and value placement of mathematical numerals, basic mathematical operations as well as measuring and describing shapes.

SUBJECT LEARNING OUTCOMES:

- 1) Demonstrate mastery of Number concepts by working out problems in day to day life.
- 2) Apply measurement skills to find solutions to problems in a variety of contexts
- 3) Describe properties of Geometrical shapes and spatial relationships in real life experiences

ISSUES IN THE REFORMED MATHEMATICS CURRICULUM INTEGRATED WITH EARLY GRADE MATHEMATICS (EGM)

Introduction

When teaching Early Grade Mathematics (EGM), it is important to integrate the following issues in the reformed curriculum designs:

- i) Core competencies to be developed
- ii) Links to pertinent and contemporary issues (PCI's)
- iii) Links to values
- iv) Links to other subjects
- v) Suggested non formal activities to support learning
- vi) Suggested community service learning activities, and
- vii) Suggested assessment

When teaching concepts in EGM it is important to refer to the information given in this handbook for guidance on how the above issues can be integrated.



1.0 STRAND: Numbers

1.1 Sub-strand: Number concept

Specific Learning Outcomes

- a) sort and group objects according to different attributes within the classroom,
- b) pair and match objects in the environment,
- c) order and sequence objects in ascending and descending order,
- d) make patterns using real objects,
- e) recite number names in order up to 50,
- f) represent numbers 1-30 using concrete objects,
- g) demonstrate through counting that a group in all situations has only one count, appreciate the use of sorting and grouping items in day to day activities

Suggested Learning Experiences

- a) In the sub-strand of the number concept, the focus will be on sorting and matching. The following learning experiences are considered: Learners in pairs/groups to collect different types of safe objects.
- b) Learners in pairs/groups to sort objects with same attribute and group them together.
- c) Learners to play digital games involving sorting and grouping according to different attributes.
- d) Learners in pairs/groups to pair and match objects to establish “equal to”, “more than” and “less than.”
- e) Learners to order objects according to size from smallest to biggest and vice versa.
- f) Learners to make patterns using real objects.
- g) Learners to recite number names up to 50.
- h) Learners to represent numbers 1-30 using concrete objects as well as their body parts.
- i) Learners to demonstrate that any given group has only one count.
- j) Learner in pairs/groups to collect and sort litter in the environment and put it in various groups according to an attribute of their choice and give reasons for the grouping.
- k) Learners in pairs/groups could assist in arranging, edible items like fruits, cabbages according to size and colour in the school store.
- l) Learners could visit a market for them to observe the sorting and grouping of fruits and vegetables.



In order to appreciate sorting and grouping in day to day activities, learners in groups / pairs may collect, sort litter and group it according to an attribute of their choice. They can be implored to give reasons for their grouping and this will enhance their communication and collaboration as well as critical thinking and problem solving skills. The teacher should advice on the safety of the materials to be collected. Learners could assist in collecting and sorting litter in their environment and observe how it is disposed, reused, and recycled thereby acquiring values such as responsibility and unity.

Assessment may be through oral questions and observations in the four categories indicated under assessment rubrics in the given sub- strand in the curriculum design.

1.0 Strand: Numbers

1.2 Sub Strand: Whole Numbers

Specific Learning Outcomes

- a) Count numbers forward and backward up to 100
- b) Represent numbers 1-50 using concrete objects
- c) Read and write numbers 1-50 in symbols
- d) Write numbers 1-10 in words
- e) Identify the place value of ones and tens
- f) make patterns using real objects
- g) identify missing numbers in number patterns up to 20
- h) appreciate number patterns by creating and extending patterns during play activities

Suggested Learning Experiences

In the sub-strand of whole numbers the following learning experiences are consider:

- a) Learner in pairs/groups to count in 1's and 2's up to 20 starting from any point using concrete objects as well as body parts.
- b) Learner to take turns in counting in 5's up to 50 starting from zero.
- c) Learner to take turns in counting in 10's up to 100 starting from zero.
- d) Learners in groups use the number line in counting in 1's, and 2's.



- e) Learners in groups play games that involve representing numbers 1-50 using concrete objects
- f) Learners in pairs to recite and write numbers 1-50 in symbols
- g) Learners to practice writing numbers 1-10 in words
- h) Learners to identify the place value of ones and tens.
- i) to make patterns using real objects
- j) to identify missing numbers in number patterns up to 20
- k) in pairs to create patterns with numbers up to 20 and share with other groups

In learning of whole numbers, the following core competencies may be developed: Learning to learn, Communication and collaboration, Critical thinking and problem solving and digital literacy. For example, while counting in 5s and 10s or making patterns using real objects, learners will develop critical thinking and problem solving skills. As learners discuss how to make patterns in groups, they communicate and collaborate towards the achievement of their goals.

As learners play games involving counting of concrete objects or making patterns using real objects in groups/ pairs it is important for the teacher to discuss safety of the materials. Reciting and writing numbers enhances learners' self esteem and the development of the vocabulary on numbers. Counting is applicable in other learning areas such as environment and language activities.

Through the recognition of the place value of digits using place value apparatus such as place value pockets, place value tins, Abacus among others, learners would develop their sense of unity and responsibility leading to their enhanced self-esteem. The knowledge of place value of digits improves the learners' vocabulary on numbers.

When forming number patterns using real objects, the teacher should ensure that learners observe safety. The display of learners' work helps to build their self-esteem and the desire to generate number patterns.



The concept of number patterns helps learners to appreciate their environment by recognizing the various patterns in nature and those arising from human activities.

The teachers is encouraged to use locally available materials and to ensure that the materials are safe and appropriate.

Learners may be involved in planting flowers or trees either at school or at home to reinforce counting, number patterns and place value. Learners may also assist in counting the total number of items in their homes or in the community or in arranging furniture or cutlery at home or at public functions in patterns. Assessment could be oral, written or by observation and should be guided by the assessment rubrics as stated in the curriculum designs.

1.0 Strand: Numbers

1.3 Sub Strand: Addition

Specific Learning Outcomes

- a) model addition as putting objects together,
- b) use '+' and '=' signs in writing addition sentences,
- c) add 2- single digit numbers up to a sum of 10,
- d) add 3- single digit numbers up to a sum of 10 in different contexts,
- e) add a 2- digit number to a 1- digit number without regrouping, horizontally and vertically with sum not exceeding 100,
- f) add multiples of 10 up to 100 vertically,
- g) work out missing numbers in patterns involving addition of whole numbers up to 100.

Suggested Learning Experiences

In addition of whole numbers, we have the following learning experiences:

- a) Learners in pairs/groups to put two groups of objects together and count to get the total.
- b) Learners to use '+' and '=' signs in writing addition sentences.
- c) Learners to add 2- single digit-numbers by skipping on a number line.
- d) Learners to add 2- single digit numbers using the family of 10.
- e) Learners to add 2- single digit number by counting on.
- f) Learners to add 3- single digit numbers using a number line.



- g) Learners to add 3- single digit numbers by counting on.
- h) Learners to add 3- single digit numbers using the family of 10.
- i) Learners to add a 2- digit number to a 1- digit number without regrouping horizontally and vertically with sum not exceeding 100.
- j) Learners to add multiples of 10 up to a 100 vertically.
- k) Learners to play digital games involving addition.
- l) Learners to make patterns involving addition with numbers up to 100.

Using a variety of strategies to teach the mastery of basic addition facts learners develop the following competencies: Learning to learn, Communication and collaboration, Critical thinking, problem solving and digital literacy. For example, by adding up to 3 single digit numbers using different strategies, learners master the basic addition facts hence enhancing critical thinking and problem solving.

The learners will use their body parts hence develop self- awareness and self- esteem. The teacher should discuss the safety of materials which the learners will work with. As learners work in groups they do so in unity and develop a sense of responsibility.

The concept of addition is applicable in other learning areas such as environmental, religious and language activities. For example, in counting and adding the different types of trees in the surrounding environment in order to establish other types of trees to plant. Learners may be involved in tree planting at school and at home. They may assist in arranging items in groups at home and in community functions. The teacher could assess the learners through oral questions, written exercises as well as observation following the assessment rubrics in the curriculum design.



1.0 Strand: Numbers

1.4 Sub-strand: Subtraction

Specific Learning Outcomes

- a) model subtraction as 'taking away' using concrete objects,
- b) use the ' - ' and '=' signs in writing subtraction sentences,
- c) subtract single digit numbers,
- d) subtract a 1- digit number from a 2- digit number based on basic addition facts,
- e) use the relationship between addition and subtraction in working out problems involving basic addition facts,
- f) subtract multiples of 10 up to 90,
- g) work out missing numbers in patterns involving subtraction of whole numbers up to 100

Suggested Learning Experiences

In this sub-strand we have the following learning experiences:

- a) Learners in pairs/groups to model subtraction using concrete objects.
- b) Learners to use ' - ' and '=' signs in writing subtraction sentences.
- c) Learners in pairs/groups to subtract by counting backwards
- d) Learners in pairs/groups to subtract using the number line.
- e) Learners to solve routine and non- routine problems involving subtraction of a 1-digit number from a 2- digit number based on basic addition facts.
- f) Learners to create subtraction sentences related to basic addition facts.
- g) Learners to use tablets to workout subtraction of multiples of 10 up to 90.
- h) Learners in pairs /groups to create patterns involving subtraction

Using a variety of strategies to teach the mastery of basic subtraction facts, learners develop the competencies of Learning to learn, Communication and collaboration, Critical thinking and problem solving and digital literacy. For example, learners will develop their critical thinking and problem solving skills using non-routine subtraction approaches. The teacher should emphasize on safety of objects used by learners. As learners work in groups they do so in unity and develop a sense of responsibility.



The knowledge of subtraction is closely linked to the learning areas of environmental and language activities. Under Environmental activities, children would be encouraged to obtain sticks from cut branches and not destroying live plants. Vocabulary is developed as learners master the basic subtraction facts using such words as ‘taking away’.

In Community Service Learning (CSL) learners would be involved collecting litter from the environment. Assessment could be undertaken through oral questions, written exercises and observations made in reference to the assessment rubrics in the curriculum designs.

2.0 Strand: Measurement

2.1 Sub Strand: Length

Specific Learning Outcomes

- a) Compare the length of objects directly
- b) Conserve length through manipulation
- c) Measure length using arbitrary units

Suggested learning experiences

Under the sub strand length, we have the following learning experiences:

- a) Learners in pairs/groups to compare objects directly to identify objects which are longer than, shorter than or same as.
- b) Learners to place objects of equal length in different orientations and describe them using words such as longer than, shorter than and same as.
- c) Learners in pairs /groups to measure lengths using different objects as arbitrary units and discuss the measurements from the various groups

When learners work in pairs/groups to compare lengths of objects and use arbitrary lengths to measure lengths of various objects they develop competences like communication and collaboration, imagination and creativity, critical thinking and problem solving and self efficacy. The teacher is expected to advice learners to be cautious when handling concrete objects to avoid unnecessary injuries. It is equally important to encourage learners to wash their hands after handling objects. The core-values of responsibility, integrity and unity are enhanced when



learners pair or work in groups towards the set goals. For example, the value of responsibility is developed when learners take charge of group activities.

The concept of length is closely linked to the learning areas like Environmental and Language activities. Under Environmental activities, children would be encouraged to obtain sticks from cut branches and not destroying live plants. The use of attributes in describing length enhances language vocabulary development.

In Community Service Learning (CSL) learners would be involved in activities such as: planting trees or flowers using sticks in determining the distance between seedlings, in religious institutions, health institutions and at home. Planting of trees or flowers in school by way of equal spaces encourages the non-formal learning dimension. Assessment could be undertaken through oral questions, written exercises and observations made in reference to the assessment rubrics in the curriculum designs.

2.0 Strand: Measurement

2.2 Sub Strand: Time

Specific Learning Outcomes

- a) Relate daily activities to time
- b) Relate days of the week with various activities

Suggested Learning Experiences

The following learning experiences are proposed under the concept of time, that is:

- a) Learners in pairs/groups to identify activities they do in the morning, afternoon and evening both at home and school.
- b) Learners to sing songs/ rhymes related to days of the week.
- c) Learners in pairs/groups to identify activities that take place during the days of the week.

By working in pairs/groups in identifying activities done at certain times at home and in school, and also when singing songs/rhymes, learners are able to develop communication and collaboration skills, self-efficacy and citizenship. In particular, the singing of the national anthem reinforces patriotism, which is a quality of good citizenship. Some activities done in certain times of the day like brushing teeth, bathing, washing clothes, cleaning generally contribute to the reduction incidences of spreading communicable diseases under health education. The



singing of the national anthem calls for respect while raising the flag, responsibility to the nation and loyalty to our nation.

The learning and knowledge of time is closely linked to such learning areas as Environmental and Languages Activities. Cleaning contributes to knowledge gained under Environmental Activities while Language is developed through the vocabulary gained relating days of the week and the accompanying activities.

For Community Service Learning, (CSL), learners would be encouraged to visit or help the needy in society during school holidays. The learners would engage in such activities as harvesting, fetching water and firewood. Non-formal dimension would be brought out when learners are encouraged to recite school daily activities during assembly sessions. Assessment could be undertaken through oral questions, written exercises and observations made in reference to the assessment rubrics in the curriculum designs.

3.0 Strand: Geometry

3.1 Sub Strand: Shapes.

Specific Learning Outcome

- a) identify rectangles, circles and triangles in the environment,
- b) make patterns involving rectangles, circles and triangles,
- c) appreciate the beauty of patterns in the environment.

Suggested Learning Experiences

- a) Learners in pairs/groups to sort and group different shapes using one attribute.
- b) Learners in pairs /groups discuss the types of lines that make rectangles, circles, triangles and name them.
- c) Learners working individually to make patterns of their choice using the three shapes.
- d) Learners in groups make patterns, colour them and share with other groups.

In learning about shapes, the following competencies may be developed: Learning to learn, communication and collaboration as well as critical thinking and problem solving. For example, while sorting and grouping the different shapes using one attribute learners will develop critical



thinking and problem solving skills. It is intended that learners will acquire a sense of responsibility, respect and work well towards the unity of the group. The concept of shapes is applicable in other learning areas such as creative art, environmental and language activities. Learners may be involved in planting flowers or trees at school or homes to reinforce learning through the use of circular, rectangular or triangular patterns. During school and community functions learners may present songs or poems while standing in circular, rectangular or triangular pattern formations.

Teachers could assess learners through written exercises, observations or oral questions as guided by the assessment rubrics in the curriculum design.

TERM TWO

No	Strand	Sub-strand
1	Numbers	Whole numbers
2	Geometry	Lines
		Shapes
3	Measurement	Length
		Capacity
		Mass

1.0 Strand: Numbers

1.1Sub Strand: Whole Numbers

- a) count numbers forward and backward up to 100,
- b) represent numbers 1-50 using concrete objects,
- c) identify place value of ones and tens,
- d) read and write numbers 1-50 in symbols,
- e) write numbers 1-10 in words,
- f) identify missing numbers in number patterns up to 20,
- g) appreciate number patterns by creating and extending patterns during play activities



Suggested learning experiences

Under whole numbers we have the following learning experiences:

- a) Learners in pairs/groups to count by 1's and 2's up to 20 starting from any point using concrete objects as well as body parts.
- b) Learners to take turns in counting by:
 - a. -5's up to 50 starting from zero
 - b. -10's up to 100 starting from zero.
- c) Learners in pairs/groups to count by 1's and 2's using a number line.
- d) Learners in pairs/groups to play games that involve representing numbers 1-50 using concrete objects.
- e) Learners to identify place value of ones and tens.
- f) Learners in pairs to recite and write numbers 1-50 in symbols.
- g) Learners to practice writing numbers 1-10 in words.
- h) Learners to identify missing numbers in number patterns up to 20.
- i) Learners in pairs to create patterns with numbers up to 20 and share with other groups.
- j) Learners to play digital games involving whole numbers.
- k) Learners to role play a cashier in day to day life activities such as a cashier counting 5 shilling coins.

As learners identify missing numbers in number patterns up to 20 and forming number families, they develop communication and collaboration skills, learning to learn, critical thinking, problem solving as well as digital literacy. For example, learners develop their critical and problem solving skills by identifying the required numbers in forming **number families**. When learners identify missing numbers and forming number families, their self esteem is enhanced. Learning number families is important in appreciating social cohesion for good citizenship. The core values of responsibility and unity are developed when learners work in groups; fully participate in the identification of missing numbers and forming number families.

The knowledge of number patterns and number families is also linked to language and religious activity areas. For example, vocabulary is developed as they identify the starting point, missing and the last numbers in the pattern.



Community Service Learning (CSL) would be enhanced by putting objects in groups of 5s, 10s together in community activities like setting dining tables and arranging chairs for different social functions. Learners may be involved in counting the number of flowers in the school compound as non-formal activities to enhance learning. Assessment may be done through oral questions, written exercises and observations using the guidance provided under the rubrics in the curriculum designs.

1.0 Strand: Numbers

1.2 Sub Strand: Addition

Learning outcomes

By the end of the sub-strand, the learner should be able to:

- a) model addition as putting objects together,
- b) use '+' and '=' signs in writing addition sentences,
- c) add 2- single digit numbers up to a sum of 10,
- d) add 3- single digit numbers up to a sum of 10 in different contexts,
- e) add a 2- digit number to a 1- digit number without regrouping, horizontally and vertically with sum not exceeding 100,
- f) add multiples of 10 up to 100 vertically,
- g) work out missing numbers in patterns involving addition of whole numbers up to 100.

Suggested learning experiences

Under addition, the following learning experiences would be considered:

- a) Learners in pairs/groups to put two groups of objects together and count to get the total.
- b) Learners to use '+' and '=' signs in writing addition sentences.
- c) Learners to add 2- single digit-numbers by skipping on a number line.
- d) Learners to add 2- single digit numbers using the family of 10.
- e) Learners to add 2- single digit number by counting on.
- f) Learners to add 3- single digit numbers using a number line.
- g) Learners to add 3- single digit numbers by counting on.
- h) Learners to add 3- single digit numbers using the family of 10.
- i) Learners to add a 2- digit number to a 1- digit number without regrouping horizontally and vertically with sum not exceeding 100.



- j) Learners to add multiples of 10 up to a 100 vertically.
- k) Learners to play digital games involving addition.
- l) Learners to make patterns involving addition with numbers up to 100.

The learning experiences of addition will enable learners to develop such competencies as critical thinking and problem solving, learning to learn, communication and collaboration, as well as digital literacy through the use of digital games. For example, after learners form the number families of 10, they could desire to explore other number families thus, developing the competence of learning to learn.

The formation of number families supports the concept of social cohesion for good citizenship. The values of integrity, responsibility and unity are also enhanced. For example, when learners use a common number line to add the given numbers, the values of unity, respect and a sense of responsibility are developed.

The concept of addition is applicable in other learning areas such as environmental, religious and language activities. For example, learners could count and add animals in a grazing field according to colour to establish their total carrying capacity.

Assessment can be done through oral questions, written exercises and observations using the guidance provided under the rubrics in the curriculum designs.

1.0 Strand: Numbers

1.3 Sub Strand: Subtraction

Learning outcome

- a) model subtraction as 'taking away' using concrete objects,
- b) use the ' - ' and '=' signs in writing subtraction sentences,
- c) subtract single digit numbers,
- d) subtract a 1- digit number from a 2- digit number based on basic addition facts,
- e) use the relationship between addition and subtraction in working out problems involving basic addition facts,
- f) subtract multiples of 10 up to 90,
- g) work out missing numbers in patterns involving subtraction of whole numbers up to 100.



Suggested learning experiences

Under subtraction, we have the following learning experiences:

- a) Learners in pairs/groups to model subtraction using concrete objects.
- b) Learners to use '-' and '=' signs in writing subtraction sentences.
- c) Learners in pairs/groups to subtract by counting backwards
- d) Learners in pairs/groups to subtract using the number line.
- e) Learners to solve routine and non-routine problems involving subtraction of a 1-digit number from a 2-digit number based on basic addition facts.
- f) Learners to create subtraction sentences related to basic addition facts.
- g) Learners to use tablets to work out subtraction of multiples of 10 up to 90.
- h) Learners in pairs/groups to create patterns involving subtraction.

As learners work in pairs/groups modelling subtraction as taking away, competencies such as learning to learn, communication and collaboration, critical thinking and problem solving, digital literacy through digital games, creativity and imagination, citizenship and self-efficacy are developed. For example, when learners work in groups/pairs to play games on subtraction on the number line, they collaborate and enhance their communication skills.

Through the process of expressing the mathematical relationship between addition and subtraction, learners would be able to develop competences such as communication and collaboration, critical thinking and problem solving, digital literacy through digital games, self-efficacy as well as creativity and imagination. For example, critical thinking and problem solving are realized through the reversal of numbers in an addition sentence to form a subtraction sentence.

When learners are collecting and using concrete objects, the teacher should remind learners of the need to observe safety. Through group work, learners are able to develop the values of unity and responsibility. The concept of subtraction is also linked to Environmental and Language activity areas. For example, when collecting concrete objects to use in class, they should be guided against destroying plants or insects. Community Service Learning (CSL) would be supported through the learners' initiative of collecting litter in their immediate environment.



Non-formal activities would involve learners weeding the school's flower beds. Assessment can be done through oral questions and observations using the guidance provided under the rubrics in the curriculum design.

2.0 Strand: Measurement

2.1 Sub Strand: Length

Specific Learning Outcomes

- a) compare length of objects directly,
- b) conserve length through manipulation,
- c) measure length using arbitrary units.

Suggested Learning Experiences

In teaching length at this point, the focus is on comparing length of objects directly. The sub strand has the following learning experiences:

- a) Learners in pairs/groups to compare objects directly to identify objects which are longer than, shorter than or same as.
- b) Learners to place objects of equal length in different orientations and describe them using words such as longer than, shorter than and same as.
- c) Learners in pairs /groups to measure lengths using different objects as arbitrary units and discuss the measurements from the various groups.

By learners working in pairs/groups to compare and order the length of objects, they develop competences such as communication and collaboration, imagination and creativity, critical thinking and problem solving and self efficacy. For example, learners who are able to order given objects through comparison of their lengths will enhance their self efficacy. The teacher should discuss with the learners on the safety of materials to be used in the activities. As learners compare the lengths of objects, they develop vocabulary on length that they can use in day to day communication and in environmental activities, e.g. that building/tree/person is taller than or shorter than the other one.

The core-values of responsibility, integrity and unity are enhanced when learners in pairs/group compare objects of different length. For example, the value of responsibility is developed when learners take charge of group activities.



The teacher may assess the learners through oral questions, written exercises as well as observation following the assessment rubrics in the curriculum design.

2.0 Strand: Measurement

2.2 Sub Strand: capacity

Specific Learning Outcomes

- a) compare capacity of containers directly,
- b) conserve capacity through manipulation,
- c) Measure capacity using arbitrary units.

Suggested Learning Experiences

In teaching capacity at this point, the focus is on comparing capacity of containers directly. In this sub strand we have the following learning experiences:

- a) Learners to empty and fill water in different containers to establish which holds more, which holds less and which holds the same.
- b) Learners to identify and compare containers which holds more, less or same as.
- c) Learners to fill containers of different shapes and sizes with water then empty into others so as to establish that some containers can hold the same amount although their shapes are different.
- d) Learners to be given water, same size basins and different small containers. The learners to count the number of small containers they use to fill the basin.

Teaching comparison of capacity of different containers helps in the development of the competencies: critical thinking and problem solving, communication and collaboration, imagination and creativity, citizenship and self-efficacy. Teacher should ensure that only safe materials are used and should discuss the safety precautions with the learners as they collect safe and appropriate containers and use them to fill and empty. As learners compare the capacities of various containers, they develop vocabulary on capacity such as holds ‘more than’, holds ‘less than’ and holds ‘same as’.

Learners develop self esteem and self-awareness as they work in groups to compare capacities by filling and emptying different containers. Learners could assist in collecting discarded containers



in their environment for disposal, re-use, and recycling, thereby acquiring the values of responsibility and unity. By re-using containers from the environment in activities on comparing capacities, learners appreciate environmental conservation. This can further be enhanced by using the containers to water trees, flowers and animals in Community Service Learning Activities.

The teacher may assess the learners through oral questions, written exercises as well as observation following the assessment rubrics in the curriculum design.

2.0 Strand: Measurement

2.3 Sub-strand: Mass

Specific Learning Outcomes

- a) compare mass of objects directly,
- b) conserve mass through manipulation,
- c) Measure mass using arbitrary units.

Suggested Learning Experiences

In teaching Mass at this point, the focus is on comparing mass of objects directly. The following learning experiences are considered:

- a) Learners in pairs/groups use safe objects to identify those heavier than, lighter than or same.
- b) Learners to use two objects of equal mass and a beam balance to demonstrate that change of shape does not change the mass of an object.
- c) Learners in pairs/groups to use an identified mass to compare the mass of other objects using the words heavier than, lighter than or same as.

Teaching comparison of mass of different objects helps in the development of the competencies: critical thinking and problem solving, communication and collaboration, imagination and creativity, citizenship and self-efficacy. For example, the self efficacy of learners is enhanced when they are able to compare the masses of different objects directly. The teacher should emphasize on safety as learners collect objects for use. Learners develop vocabulary such as ‘heavier than’, ‘lighter than’ or ‘same as’ they compare masses of different objects.



Learners develop self esteem and self-awareness when they are able to compare masses of different objects. Learners could assist in collecting discarded objects in their environment for disposal, re-use, and recycling, thereby acquiring the values of responsibility and unity. By re-using objects from the environment in activities on comparing mass, learners appreciate environmental conservation. Learners could develop their non-formal dimension by compare masses of objects in the classroom or within the school.

For Community Service Learning, learners could assist parents and community in feeding animals by measuring quantities.

The teacher may assess the learners through oral questions, written exercises as well as observation following the assessment rubrics in the curriculum design.

3.0 STRAND: Geometry

3.1 Sub-strand: Shapes.

Specific Learning Outcomes

- a) Identify rectangles, circles and triangles in the environment.
- b) Make patterns involving rectangles circles and triangles.
- c) Appreciate the beauty of patterns in the environment.

Suggested Learning Experiences

In this sub-strand we have the following learning experiences:

- a) Learners in pairs/groups to sort and group different shapes using one attribute.
- b) Learners in pairs/groups discuss the different types of shapes and name them.
- c) Learners working individually to make patterns of their choice using the three shapes.
- d) Learners in groups make patterns, colour them and share with other groups.

The teacher should involve the learners in sorting and grouping different shapes in groups. Creativity and imagination of learners is developed as they sort and group different shapes using given attributes. During the discussions on the different shapes, learners develop collaboration and communication competencies. As the learners trace and colour patterns, the teacher should



discuss the need for safety. When learners are involved in sorting and grouping shapes, they develop a sense of self- awareness and self- esteem. Values such as unity and respect are developed as learners work in groups.

Learners could visit pre-school and decorate the walls using the different patterns learned. Use of shapes in pattern making is applicable in movement and creative arts. Learners could visit homes for the elderly to decorate the walls with patterns drawn on Manila paper. They could also visit places of worship to assist in decorating using different patterns of the three shapes. It is suggested that teachers can assess the learners through written exercises, observations or oral questions as guided by the assessment rubrics in the curriculum design.

3.0 Strand: Geometry

3.2 sub Strand: Lines

Specific learning outcomes

- a) Draw straight lines for application in real life,
- b) Draw curved lines for application in real life situations.

Suggested learning experiences

In this sub-strand we have the following learning experiences:

- a) Learners to stand behind one another facing the same side and identify what they have formed as a straight line.
- b) Learners in pairs/groups to mark two points on the ground and using a stick to draw a line joining the two points to come up with a straight line.
- c) Learners to practice drawing straight lines on the ground and in their books.
- d) Learners in groups to form a semi-circle and one of them to draw a line around it and identify the semi-circle drawn as a curved line.
- e) Learners to practice drawing curved lines on the ground and in their books.
- f) Learners could visit a water selling kiosk to observe how the water containers are arranged.

The teacher should involve learners in identifying and drawing straight and curved lines in groups. Their imagination and creativity skills will be developed in the process of identification



and drawing on the ground and in their books. The teacher should guide learners on the safety of the materials they use in drawing. As learners draw lines on the ground and in their books, they develop a sense self-awareness and self-esteem. The concept of lines is also applicable in creative arts. Learners may be asked to plant trees and flowers on straight and curved formations. Learners may be involved in arranging seats in straight and curved formations for social functions. It is suggested that teacher can assess learners through observation, written exercises and oral questions as guided by the assessment rubrics in the curriculum design.

School Report Card

School

Grade.....Term.....

Name of Learner

Learning Area	CRITERIA (<i>Based on Expected Outcomes in area curriculum design</i>)	Narrative Comments by Learning Area teacher (Based on Rubrics)
Mathematics Activities	• Subtraction of single and two digit numbers • Number patterns	Peter correctly uses subtractions sign (-) to write subtraction sentences. He correctly subtracts using concrete objects and also subtracts up to two digit numbers. He is able to work out missing numbers in number patterns up to 100 correctly.
Environmental Activities		



Language Activities		
Music and movement activities		

CORE COMPETENCES (The teacher to make comments on observations during the term on the learners abilities in the following areas)	
Community Service Learning	
Communication and collaboration Digital literacy	
Critical thinking and problem solving	
Creativity and imagination	
Citizenship	
Any other Comments from other teachers	
Comments by School Head Teacher (To feature Strengths and Targets)	



Comments by Parent/Guardian	
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KARANDI PRIMARY SCHOOL

Report of achievement TERM 3 2017 GRADE 2

LEARNER: PETER AMBILA

English Activities:

Peter reads fluently and with excellent comprehension. He is an articulate speaker with a wide oral vocabulary. Peter can ride in a range of contexts with a good awareness of vocabulary, structure and punctuation.

Maths Activities:

Peter has excellent mental and written maths skills. He can select and use appropriate operations in working out solutions to problem. Peter is able to recall multiplication facts reliably.

Environmental Activities:

Peter has a good understanding of properties and reactions. He is able to predict outcomes and record his understanding clearly. Peter enjoys investigations and he is able to explain his thinking with reasoned logic.

Creative Art Activities:

Peter has worked in a range of media, at varying scales. He is developing a detailed observational drawing style.

Music and movement Activities:

Peter is able to play simple tunes on the recorder. He can follow basic musical notation reliably. Peter can identify rhythmic patterns in music and replicate this using percussion.

Physical Education Activities



Peter has developed good team skills in football, Hockey and Rugby. He can sequence movement in dance. James enjoys swimming and able to evaluate performance.

Religious Activities : Peter has a good understanding of the Christian ethos. He can locate books in Bible and is able to recall stories with good details. Peter has learned about the lives of many different saints and he understandings their contributions to our lives.

Strengths and Targets

Peter has a conscientious attitude to his work. He strives to do his best. He is an active learner with advanced thinking skills. Peter should aim to develop the presentation of his work and to continue to use vocabulary and structure to maintain the quality of his writing.

Signed : _____

Class Teacher

Head

Date